

Ongoing Work on Contextual Neural Machine Translation

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Motivation

State of the Art Literature

Ongoing Work at i6

Experiments

Summary and Future Work

- ▶ Most neural machine translation (NMT) systems translate sentence by sentence
- ▶ Make use of other sentences in addition to exploit context information
 - ▷ reduce ambiguities in translation choices
 - ▷ example: cross-sentence anaphora [Voita & Sennrich⁺ 18]

An English→German example of a cross-sentence anaphora problem. The translation of the pronoun '**it**' depends on the previous sentence.

Source sentences	Correct translation of ' it '
<i>The monkey ate the banana. Because it was hungry.</i>	er
<i>The monkey ate the banana. Because it ripe.</i>	sie
<i>The monkey ate the banana. Because it was tea-time.</i>	es

- ▶ **Single encoder: concatenate sentences on source-/target side**
[Tiedemann & Scherrer 17, Agrawal & Turchi⁺ 18]

- ▶ **Multi-Encoder with attention mechanism**
 - ▷ **Local context: feed surrounding sentences into separate encoders. Combine context vectors:**
 - concatenation (like in [Zoph & Knight 16])
 - gated sum [Bawden & Sennrich⁺ 18]
 - hierarchical attention [Bawden & Sennrich⁺ 18]

Remark: above methods only consider *local* context

1. Reproduce results from [Agrawal & Turchi⁺ 18]

- ▶ using Transformer model as well
- ▶ first, find good baseline: hyperparameter search
 - ▷ weight-tying
 - ▷ learning rates
 - ▷ checkpoints
 - ▷ ...
- ▶ concatenate sentences with special token **<BREAK>**,
example: sentence_1 **<BREAK>** sentence_2

Ongoing Work at i6 - Concatenation of sentences

(a) Single target sentence.

Source input	Target input
1 previous + current	current
2 previous + current	current
1 previous + current + 1 next	current
current + 1 next	current

(b) Multiple target sentence.

Source input	Target input
1 previous + current	1 previous + current
1 previous + current + 1 next	1 previous + current

Experiments

Selected data sets for experiments.

	IWSLT'17 EN-IT	WMT'18 DE-EN
Train.	IWSLT'17	WMT'18
Dev.	<i>tst2010</i>	<i>newstest2016</i>
Test	<i>tst2017</i>	<i>newstest2017</i>

► English→Italian:

Experiments: Data set statistics

Corpus statistics for both IWSLT 2017 English→Italian and WMT 2018 German→English data sets.

		IWSLT'17 EN-IT		WMT'18 DE-EN	
		EN	IT	DE	EN
Train.	Sentences	232K		5.9M	
	Running words	4.7M	4.4M	160.9M	157.7M
Dev.	Sentences	1.5K		3K	
	Running words	31K	29K	77K	73K
Test	Sentences	1.1K		3K	
	Running words	22K	20K	76K	73K

Thank you for your attention

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